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*Environmental Regulation and Green Skills: An Empirical Exploration*

Journal of the Association of Environmental and Resource Economists

**SUPPLEMENTARY MATERIAL FOR ONLINE PUBLICATION ONLY**

# Web Appendix A - Green Skills

This appendix provides details on the data source and the procedure for the selection of *GGS* based on the greenness of green occupations (the full list of green occupations and their level of greenness is in Table A1). Table A2 reports the estimated $\beta$ of equation 2 for all general skills and tasks for which the beta was positive and significant at the 99% level or more. Recall that results are based on 475 occupations observed at the 8-digit SOC level for the year 2012 (3-digit SOC occupations without any green jobs were excluded, see Table A3 for the list of 3-digit SOC occupations with at least one green job and Table A4 for the list of 3-digit SOC occupations without any green job) and regressions include 3-digit SOC dummies. Out of 108 general skills and tasks, 16 have been selected as particularly relevant for green occupations.

As discussed in section 2.2, we perform a principal component analysis on these 20 general skills and tasks to generate more aggregate measures of *GGS* importance. As discussed in section 3.2, we retain five components with Eigenvalues (unrotated components) respectively 5.58, 3.93, 1.34, 0.99 and 0.92, and a cumulative explained variance of 79.72 percent. Table A5 shows the factor loadings of the 5 rotated components (orthogonal VARIMAX rotation) that exceeded a 0.2 threshold. The first groups together "Engineering & Technical Skills". The second component, labelled "Operation Management Skills", includes abilities that are relevant for management practices associated with new technology. The third component is "Monitoring Skills". Therein we observe that two general skills (Law and Government and Evaluating Information to Determine Compliance with Standards) load much more than the third (Operating Vehicles, Mechanized Devices, or Equipment) which, instead, loads negatively on the second component. A thorough reading of the description of these skills (from O*NET) reveals that only the first two bear direct relevance for Monitoring activities, while the third one has to do with operating machineries, vehicles and means of transport and thus not only with the use of monitoring devices. We therefore excluded this item. The fourth component clearly refers to Science Skills. Finally, the fifth component is characterized by a large factor loading (Geography, 0.84) and a smaller loading one (Law and Government which, however, was already assigned to component 3). Geographic skills capture activities such as urban planning and analysis of emission dynamics (several professions intensive of Geography skills are green, such as Environmental Restoration Planners, Landscape Architects and Atmospheric and Space Scientist). Due to the highly specific nature we opted to exclude this component. Results on the impact of environmental regulation for this *GGS* and for each single general skill selected here (including "Geography" and "Operating Vehicles, Mechanized Devices, or Equipment", which were excluded from the *GGS* constructs) are shown in the online supplementary Appendix D.

We tried several alternative methods of selecting *GGS* to assess the robustness of our selection procedure and to identify *GGS* that are selected irrespective of the procedure. We present here three of these additional trials. First, as already discussed in section 2.2, we only focus on green tasks that are central to the sustainability transition. Accordingly, we searched for keywords in the description of the green specific tasks that helped us to identify the most appropriate tasks. The full list of keywords is reported in Table A6. This method led to the exclusion of 160 of the 1373 green specific tasks, or 11.65 percent. We repeat the main data-driven exercise to select *GGS* on a modified greenness that just considers green specific tasks selected by keyword search. Results are reported in Table A7. The new procedure selects 15 items, 13 of which overlap with those selected through our preferred approach. All 'Engineering and Technical' and 'Science' *GGS* items are selected, thus confirming the importance of technical and scientific skills. In the 'Monitoring' group, 'Law and Government' was no longer selected, while 'Provide Consultation and Advice to Others' was no longer selected in the group 'Operation Management'. The residual item ' Operating Vehicles, Mechanized Devices, or Equipment' was no longer selected while two new items were selected: 'Transportation' (that is very much correlated with 'Geography') and 'Developing Objectives and Strategies' that enters the group 'Operation Management'.

Second, we estimate equation 2 by weighting each occupation by the total of employees in year 2012[1]. Note that this is not our favourite method because it assigns undue weight to occupations that are overrepresented in the service sector and, thus, not directly affected by sustainability issues. Results are reported in Table A8. This method only retains general skills that enter two of our Engineering & Technical and Science skills constructs, with the addition of Chemistry that was not selected in our preferred approach. Engineering & Technical and Science skills encompass the core technical and scientific know-how that is required in green occupations.

Finally, we decompose *Greenness* into two components, namely, the count of green specific tasks and the count of total specific tasks. Under this specification we allow both components of the *Greenness* indicator to have an independent effect on general skills. Results for the coefficients associated with green specific tasks and total specific tasks are reported in Table A9 in the online supplementary Appendix A. Therein we observe a positive and significant (at the 99 percent level) relationship between the number of green specific tasks for 13 general skills. Out of these 13 skills only *Systems Evaluation* shows a positive and significant correlation with the total number of specific tasks. These 13 items are a subset of our initial selection of 16

[1] Weights at the 6-digit SOC level for year 2012 are based on the Occupational Employment Statistics prepared by the Bureau of Labor Statistics. It collects, among other things, aggregate employment measures by detailed occupation. No information is available at the 8-digit SOC level. As discussed in online supplementary Appendix B about state-industry measures, we decide to weight equally each 8-digit occupation within its corresponding 6-digit macro-occupation.

general skills. This second criterion excludes two general skills that entered the Operation Management *GGS* (System Analysis and Updating and Using Relevant Knowledge) and one Science skills (Biology).

Taking the cue from the polarization of occupations within engineering skills, in **Error! Reference source not found.** we take a closer look at the component parts of this construct: Engineering & Technology, Design, Building & Construction, Mechanical, Drafting and Estimating quantifiable characteristics. The descriptions provided by O*NET offer a point of reference for detecting functional commonalities and differences across these items. Engineering & Technology and Design are areas of knowledge associated with the application of scientific principles to practical problems. By contrast, Building & Construction, Mechanical, Drafting and Estimating quantifiable characteristics pertain to areas of practical know-how of e.g. materials, machines, tools as well as of the technical specifications that are necessary to operate them. In short, the first two items of Green Engineering skills are about "Conceiving solutions" while the remaining three are about "Implementing solutions". This functional difference is reflected also in the educational levels associated with each of these specific skills. The upper portion of **Error! Reference source not found.** shows that an average 21% of workers in occupations with the highest (top 10%) value of Engineering and Technical skills possess a college degree, while only 5% have postgraduate education. At the same time, the high standard deviation for college graduates suggests strong within group variability, confirmed also by the mean values of each individual skill item. In particular, Engineering & Technology and Design look rather similar since for both the average number of top occupations with at least college degree is above 40%. This is not so for the other items, in particular for Mechanical and Drafting skills where average values range between 8% and 18% respectively. Such a polarization in the educational requirements of occupations with the highest intensity of green Engineering skills hints at interesting heterogeneity in the type of knowledge possessed by these workers with vocational and technical degrees more important for "low" engineering skills.

Table A1 - List of jobs using green skills

| SOC 2010 | Title | *Greenness* | Total spec tasks | Green spec tasks |
|---|---|---|---|---|
| 11-1011.03 | Chief Sustainability Officers | 1.00 | 18 | 18 |
| 11-1021.00 | General and Operations Managers | 0.06 | 18 | 1 |
| 11-2021.00 | Marketing Managers | 0.20 | 20 | 4 |
| 11-3051.02 | Geothermal Production Managers | 1.00 | 17 | 17 |
| 11-3051.04 | Biomass Power Plant Managers | 1.00 | 18 | 18 |
| 11-3071.01 | Transportation Managers | 0.18 | 28 | 5 |
| 11-3071.02 | Storage and Distribution Managers | 0.23 | 30 | 7 |
| 11-3071.03 | Logistics Managers | 0.30 | 30 | 9 |
| 11-9021.00 | Construction Managers | 0.28 | 25 | 7 |
| 11-9041.00 | Architectural and Engineering Managers | 0.19 | 21 | 4 |
| 11-9121.02 | Water Resource Specialists | 1.00 | 21 | 21 |
| 11-9199.01 | Regulatory Affairs Managers | 0.15 | 27 | 4 |
| 11-9199.02 | Compliance Managers | 0.20 | 30 | 6 |
| 11-9199.04 | Supply Chain Managers | 0.30 | 30 | 9 |
| 11-9199.11 | Brownfield Redevelopment Specialists and Site Managers | 1.00 | 22 | 22 |
| 13-1022.00 | Wholesale and Retail Buyers, Except Farm Products | 0.24 | 21 | 5 |
| 13-1041.07 | Regulatory Affairs Specialists | 0.19 | 32 | 6 |
| 13-1081.01 | Logistics Engineers | 0.37 | 30 | 11 |
| 13-1081.02 | Logistics Analysts | 0.19 | 31 | 6 |
| 13-1151.00 | Training and Development Specialists | 0.10 | 21 | 2 |
| 13-1199.01 | Energy Auditors | 1.00 | 21 | 21 |
| 13-1199.05 | Sustainability Specialists | 1.00 | 14 | 14 |
| 13-2051.00 | Financial Analysts | 0.33 | 18 | 6 |
| 13-2052.00 | Personal Financial Advisors | 0.14 | 21 | 3 |
| 13-2099.02 | Risk Management Specialists | 0.17 | 24 | 4 |
| 15-1199.04 | Geospatial Information Scientists and Technologists | 0.08 | 24 | 2 |
| 15-1199.05 | Geographic Information Systems Technicians | 0.26 | 19 | 5 |
| 17-1011.00 | Architects, Except Landscape and Naval | 0.37 | 19 | 7 |
| 17-1012.00 | Landscape Architects | 0.26 | 19 | 5 |
| 17-2011.00 | Aerospace Engineers | 0.33 | 18 | 6 |
| 17-2051.00 | Civil Engineers | 0.47 | 17 | 8 |
| 17-2051.01 | Transportation Engineers | 0.23 | 26 | 6 |
| 17-2071.00 | Electrical Engineers | 0.14 | 22 | 3 |
| 17-2072.00 | Electronics Engineers, Except Computer | 0.22 | 23 | 5 |
| 17-2081.00 | Environmental Engineers | 1.00 | 28 | 28 |
| 17-2081.01 | Water/Wastewater Engineers | 1.00 | 27 | 27 |
| 17-2141.00 | Mechanical Engineers | 0.26 | 27 | 7 |
| 17-2161.00 | Nuclear Engineers | 0.35 | 20 | 7 |
| 17-2199.01 | Biochemical Engineers | 0.34 | 35 | 12 |
| 17-2199.02 | Validation Engineers | 0.09 | 22 | 2 |
| 17-2199.03 | Energy Engineers | 0.95 | 21 | 20 |
| 17-2199.04 | Manufacturing Engineers | 0.17 | 24 | 4 |
| 17-2199.05 | Mechatronics Engineers | 0.13 | 23 | 3 |
| 17-2199.07 | Photonics Engineers | 0.19 | 26 | 5 |
| 17-2199.08 | Robotics Engineers | 0.08 | 24 | 2 |
| 17-2199.10 | Wind Energy Engineers | 1.00 | 16 | 16 |
| 17-3023.03 | Electrical Engineering Technicians | 0.21 | 24 | 5 |
| 17-3024.00 | Electro-Mechanical Technicians | 0.08 | 12 | 1 |
| 17-3024.01 | Robotics Technicians | 0.09 | 23 | 2 |
| 17-3025.00 | Environmental Engineering Technicians | 1.00 | 26 | 26 |
| 17-3026.00 | Industrial Engineering Technicians | 0.22 | 18 | 4 |
| 17-3029.02 | Electrical Engineering Technologists | 0.40 | 20 | 8 |
| 17-3029.03 | Electromechanical Engineering Technologists | 0.29 | 17 | 5 |
| 17-3029.04 | Electronics Engineering Technologists | 0.17 | 23 | 4 |
| 17-3029.05 | Industrial Engineering Technologists | 0.17 | 23 | 4 |
| 17-3029.06 | Manufacturing Engineering Technologists | 0.28 | 29 | 8 |
| 17-3029.07 | Mechanical Engineering Technologists | 0.14 | 21 | 3 |
| 17-3029.08 | Photonics Technicians | 0.20 | 30 | 6 |
| 17-3029.09 | Manufacturing Production Technicians | 0.20 | 30 | 6 |
| 19-1013.00 | Soil and Plant Scientists | 0.63 | 27 | 17 |
| 19-1031.01 | Soil and Water Conservationists | 1.00 | 33 | 33 |
| 19-2021.00 | Atmospheric and Space Scientists | 0.50 | 24 | 12 |
| 19-2041.01 | Climate Change Analysts | 1.00 | 14 | 14 |
| 19-2041.02 | Environmental Restoration Planners | 1.00 | 22 | 22 |
| 19-2042.00 | Geoscientists, Except Hydrologists and Geographers | 0.48 | 31 | 15 |
| 19-2099.01 | Remote Sensing Scientists and Technologists | 0.08 | 24 | 2 |
| 19-3011.01 | Environmental Economists | 1.00 | 19 | 19 |

| SOC 2010 | Title | *Greenness* | Total spec tasks | Green spec tasks |
|---|---|---|---|---|
| 19-3051.00 | Urban and Regional Planners | 0.37 | 19 | 7 |
| 19-3099.01 | Transportation Planners | 0.14 | 22 | 3 |
| 19-4011.01 | Agricultural Technicians | 0.12 | 25 | 3 |
| 19-4041.01 | Geophysical Data Technicians | 0.24 | 21 | 5 |
| 19-4041.02 | Geological Sample Test Technicians | 0.19 | 16 | 3 |
| 19-4051.01 | Nuclear Equipment Operation Technicians | 0.41 | 17 | 7 |
| 19-4091.00 | Environmental Science and Protection Technicians, Including Health | 1.00 | 25 | 25 |
| 19-4099.02 | Precision Agriculture Technicians | 0.30 | 23 | 7 |
| 19-4099.03 | Remote Sensing Technicians | 0.14 | 22 | 3 |
| 23-1022.00 | Arbitrators, Mediators, and Conciliators | 0.05 | 20 | 1 |
| 27-3022.00 | Reporters and Correspondents | 0.05 | 22 | 1 |
| 27-3031.00 | Public Relations Specialists | 0.24 | 17 | 4 |
| 29-9012.00 | Occupational Health and Safety Technicians | 0.35 | 26 | 9 |
| 41-4011.00 | Sales Representatives, Wholesale and Manufacturing, Technical and Scientific Products | 0.11 | 38 | 4 |
| 41-4011.07 | Solar Sales Representatives and Assessors | 1.00 | 13 | 13 |
| 43-5071.00 | Shipping, Receiving, and Traffic Clerks | 0.09 | 11 | 1 |
| 47-2061.00 | Construction Laborers | 0.18 | 33 | 6 |
| 47-2152.01 | Pipe Fitters and Steamfitters | 0.15 | 20 | 3 |
| 47-2152.02 | Plumbers | 0.39 | 23 | 9 |
| 47-2181.00 | Roofers | 0.30 | 30 | 9 |
| 47-2211.00 | Sheet Metal Workers | 0.24 | 25 | 6 |
| 47-2231.00 | Solar Photovoltaic Installers | 1.00 | 26 | 26 |
| 47-4011.00 | Construction and Building Inspectors | 0.26 | 19 | 5 |
| 47-4041.00 | Hazardous Materials Removal Workers | 0.91 | 23 | 21 |
| 47-4099.03 | Weatherization Installers and Technicians | 1.00 | 18 | 18 |
| 47-5013.00 | Service Unit Operators, Oil, Gas, and Mining | 0.05 | 19 | 1 |
| 47-5041.00 | Continuous Mining Machine Operators | 0.17 | 12 | 2 |
| 49-3023.02 | Automotive Specialty Technicians | 0.40 | 25 | 10 |
| 49-3031.00 | Bus and Truck Mechanics and Diesel Engine Specialists | 0.16 | 25 | 4 |
| 49-9021.01 | Heating and Air Conditioning Mechanics and Installers | 0.23 | 30 | 7 |
| 49-9071.00 | Maintenance and Repair Workers, General | 0.13 | 31 | 4 |
| 49-9081.00 | Wind Turbine Service Technicians | 1.00 | 13 | 13 |
| 49-9099.01 | Geothermal Technicians | 1.00 | 24 | 24 |
| 51-2011.00 | Aircraft Structure, Surfaces, Rigging, and Systems Assemblers | 0.13 | 30 | 4 |
| 51-4041.00 | Machinists | 0.07 | 29 | 2 |
| 51-8011.00 | Nuclear Power Reactor Operators | 0.33 | 18 | 6 |
| 51-8013.00 | Power Plant Operators | 0.21 | 24 | 5 |
| 51-8099.03 | Biomass Plant Technicians | 1.00 | 16 | 16 |
| 51-9012.00 | Separating, Filtering, Clarifying, Precipitating, and Still Machine Setters, Operators, and Tenders | 0.05 | 20 | 1 |
| 51-9061.00 | Inspectors, Testers, Sorters, Samplers, and Weighers | 0.06 | 32 | 2 |
| 51-9199.01 | Recycling and Reclamation Workers | 1.00 | 18 | 18 |
| 53-3032.00 | Heavy and Tractor-Trailer Truck Drivers | 0.09 | 33 | 3 |
| 53-6051.07 | Transportation Vehicle, Equipment and Systems Inspectors, Except Aviation | 0.41 | 22 | 9 |
| 53-7081.00 | Refuse and Recyclable Material Collectors | 1.00 | 16 | 16 |

## Table A2 – Selection of green skills

| Item | Description | Beta | S.E. |
|---|---|---|---|
| 2B4g | Systems Analysis | 0.0589*** | (0.0185) |
| 2B4h | Systems Evaluation | 0.0603*** | (0.0182) |
| 2C3b | Engineering and Technology | 0.181*** | (0.0518) |
| 2C3c | Design | 0.158*** | (0.0451) |
| 2C3d | Building and Construction | 0.203*** | (0.0503) |
| 2C3e | Mechanical | 0.135*** | (0.0514) |
| 2C4b | Physics | 0.182*** | (0.0546) |
| 2C4d | Biology | 0.0933*** | (0.0301) |
| 2C4g | Geography | 0.140*** | (0.0331) |
| 2C8b | Law and Government | 0.0948*** | (0.0345) |
| 4A1b3 | Estimating the Quantifiable Characteristics of Products, Events, or Information | 0.0563*** | (0.0196) |
| 4A2a3 | Evaluating Information to Determine Compliance with Standards | 0.0553*** | (0.0185) |
| 4A2b3 | Updating and Using Relevant Knowledge | 0.0482*** | (0.0180) |
| 4A3a4 | Operating Vehicles, Mechanized Devices, or Equipment | 0.0942*** | (0.0310) |
| 4A3b2 | Drafting, Laying Out, and Specifying Technical Devices, Parts, and Equipment | 0.124*** | (0.0373) |
| 4A4b6 | Provide Consultation and Advice to Others | 0.0666*** | (0.0206) |

N=475 occupations (8-digit SOC). 3-digit SOC occupations with no green occupations are excluded. 3-digit SOC dummies included. OLS estimates. Standard errors clustered by 3-digit SOC in parenthesis. Beta and S.E. refer to the variable *Greenness*

## Table A3 – 3-digit SOC occupations that contain at least one green job

| SOC | Description |
|---|---|
| 11-1 | Top Executives |
| 11-2 | Advertising, Marketing, Promotions, Public Relations, and Sales Managers |
| 11-3 | Operations Specialties Managers |
| 11-9 | Other Management Occupations |
| 13-1 | Business Operations Specialists |
| 13-2 | Financial Specialists |
| 15-1 | Computer Occupations |
| 17-1 | Architects, Surveyors, and Cartographers |
| 17-2 | Engineers |
| 17-3 | Drafters, Engineering Technicians, and Mapping Technicians |
| 19-1 | Life Scientists |
| 19-2 | Physical Scientists |
| 19-3 | Social Scientists and Related Workers |
| 19-4 | Life, Physical, and Social Science Technicians |
| 23-1 | Lawyers, Judges, and Related Workers |
| 27-3 | Media and Communication Workers |
| 29-9 | Other Healthcare Practitioners and Technical Occupations |
| 41-4 | Sales Representatives, Wholesale and Manufacturing |
| 43-5 | Material Recording, Scheduling, Dispatching, and Distributing Workers |
| 47-2 | Construction Trades Workers |
| 47-4 | Other Construction and Related Workers |
| 47-5 | Extraction Workers |
| 49-3 | Vehicle and Mobile Equipment Mechanics, Installers, and Repairers |
| 49-9 | Other Installation, Maintenance, and Repair Occupations |
| 51-2 | Assemblers and Fabricators |
| 51-4 | Metal Workers and Plastic Workers |
| 51-8 | Plant and System Operators |
| 51-9 | Other Production Occupations |
| 53-3 | Motor Vehicle Operators |
| 53-6 | Other Transportation Workers |
| 53-7 | Material Moving Workers |

Table A4 – 3-digit SOC occupations that do not contain any green job

| SOC | Description |
|---|---|
| 15-2 | Mathematical Science Occupations |
| 21-1 | Counselors, Social Workers, and Other Community and Social Service Specialists |
| 21-2 | Religious Workers |
| 23-2 | Legal Support Workers |
| 25-1 | Postsecondary Teachers |
| 25-2 | Preschool, Primary, Secondary, and Special Education School Teachers |
| 25-3 | Other Teachers and Instructors |
| 25-4 | Librarians, Curators, and Archivists |
| 25-9 | Other Education, Training, and Library Occupations |
| 27-1 | Art and Design Workers |
| 27-2 | Entertainers and Performers, Sports and Related Workers |
| 27-4 | Media and Communication Equipment Workers |
| 29-1 | Health Diagnosing and Treating Practitioners |
| 29-2 | Health Technologists and Technicians |
| 31-1 | Nursing, Psychiatric, and Home Health Aides |
| 31-2 | Occupational Therapy and Physical Therapist Assistants and Aides |
| 31-9 | Other Healthcare Support Occupations |
| 33-1 | Supervisors of Protective Service Workers |
| 33-2 | Fire Fighting and Prevention Workers |
| 33-3 | Law Enforcement Workers |
| 33-9 | Other Protective Service Workers |
| 35-1 | Supervisors of Food Preparation and Serving Workers |
| 35-2 | Cooks and Food Preparation Workers |
| 35-3 | Food and Beverage Serving Workers |
| 35-9 | Other Food Preparation and Serving Related Workers |
| 37-1 | Supervisors of Building and Grounds Cleaning and Maintenance Workers |
| 37-2 | Building Cleaning and Pest Control Workers |
| 37-3 | Grounds Maintenance Workers |
| 39-1 | Supervisors of Personal Care and Service Workers |
| 39-2 | Animal Care and Service Workers |
| 39-3 | Entertainment Attendants and Related Workers |
| 39-4 | Funeral Service Workers |
| 39-5 | Personal Appearance Workers |
| 39-6 | Baggage Porters, Bellhops, and Concierges |
| 39-7 | Tour and Travel Guides |
| 39-9 | Other Personal Care and Service Workers |
| 41-1 | Supervisors of Sales Workers |
| 41-2 | Retail Sales Workers |
| 41-3 | Sales Representatives, Services |
| 41-9 | Other Sales and Related Workers |
| 43-1 | Supervisors of Office and Administrative Support Workers |
| 43-2 | Communications Equipment Operators |
| 43-3 | Financial Clerks |
| 43-4 | Information and Record Clerks |
| 43-6 | Secretaries and Administrative Assistants |
| 43-9 | Other Office and Administrative Support Workers |
| 45-1 | Supervisors of Farming, Fishing, and Forestry Workers |
| 45-2 | Agricultural Workers |
| 45-3 | Fishing and Hunting Workers |
| 45-4 | Forest, Conservation, and Logging Workers |
| 47-1 | Supervisors of Construction and Extraction Workers |
| 47-3 | Helpers, Construction Trades |
| 49-1 | Supervisors of Installation, Maintenance, and Repair Workers |
| 49-2 | Electrical and Electronic Equipment Mechanics, Installers, and Repairers |
| 51-1 | Supervisors of Production Workers |
| 51-3 | Food Processing Workers |
| 51-5 | Printing Workers |
| 51-6 | Textile, Apparel, and Furnishings Workers |
| 51-7 | Woodworkers |
| 53-1 | Supervisors of Transportation and Material Moving Workers |
| 53-2 | Air Transportation Workers |
| 53-4 | Rail Transportation Workers |
| 53-5 | Water Transportation Workers |

Table A5 – Principal component analysis

| Item | Description | Component 1 | Component 2 | Component 3 | Component 4 | Component 5 |
|---|---|---|---|---|---|---|
| 2B4g | Systems Analysis | | 0.4346 | | | |
| 2B4h | Systems Evaluation | | 0.4245 | | | |
| 2C3b | Engineering and Technology | 0.4278 | | | | |
| 2C3c | Design | 0.4536 | | | | |
| 2C3d | Building and Construction | 0.3021 | | | | 0.2204 |
| 2C3e | Mechanical | 0.3326 | -0.2976 | | | |
| 2C4b | Physics | 0.3191 | | | 0.4405 | |
| 2C4d | Biology | | | | 0.8000 | |
| 2C4g | Geography | | | | | 0.8432 |
| 2C8b | Law and Government | | | 0.4602 | | 0.3856 |
| 4A1b3 | Estimating the Quantifiable Characteristics of Products, Events, or Information | 0.2564 | | | | |
| 4A2a3 | Evaluating Information to Determine Compliance with Standards | | | 0.6999 | | -0.2124 |
| 4A2b3 | Updating and Using Relevant Knowledge | | 0.3241 | | | |
| 4A3a4 | Operating Vehicles, Mechanized Devices, or Equipment | | -0.5026 | 0.3407 | | |
| 4A3b2 | Drafting, Laying Out, and Specifying Technical Devices, Parts, and Equipment | 0.4298 | | | | |
| 4A4b6 | Provide Consultation and Advice to Others | | 0.3535 | 0.2250 | | |

Principal component analysis. VARIMAX rotated components with loadings<0.2 not shown. Cumulative explained variance (5 components): 79.72%. Eigenvalues for the first six unrotated components: 5.58, 3.93, 1.34, 0.99, 0.92, 0.65.

Table A6– Keywords list for the selection of core green specific tasks

| **Single words** | | | |
|---|---|---|---|
| *Array* | *Erosion* | *Natural resource* | *Sulf* |
| *Biomass* | *Fuel* | *Nitrog* | *Sustainab* |
| *Brownfield* | *Geotherm* | *Ozon* | *Test* |
| *Carbon* | *Green* | *Photovoltaic* | *Toxic* |
| *Clean* | *Hazardous* | *Pollut* | *Treat* |
| *Climat* | *Hybrid vehicle* | *Purif* | *Warm* |
| *Conserv* | *Hydro* | *Reclaim* | *Waste management* |
| *Contamin* | *Industrial ecolog* | *Recover* | *Wastewater* |
| *Ecologic* | *Insulat* | *Recycl* | *Water* |
| *Ecosystem* | *Inverter* | *Regulat* | *Weather seal* |
| *Electric* | *Land use* | *Remedi* | *Weatherization* |
| *Emission* | *Landfill gas* | *Renewable* | *Wetland* |
| *Energ* | *Mitigat* | *Restor* | *Wind* |
| *Environment* | *Monitor* | *Solar* | |
| **Boolean search** | | | |
| *Efficien* AND *heat* AND *boiler* | | | |

**Other core green specific tasks**

- *Evaluate power production or demand trends to identify opportunities for improved operations*
- *Develop site-specific health and safety protocols, such as spill contingency plans or methods for loading or transporting waste*
- *Redesign packaging for manufactured products to minimize raw material use or waste*
- *Reengineer nanomaterials to improve biodegradability*
- *Identify or recommend integrated weed and pest management (ipm) strategies, such as resistant plants, cultural or behavioral controls, soil amendments, insects, natural enemies, barriers, or pesticides*
- *Calculate or compare efficiencies associated with changing from low-precision irrigation technologies, such as furrow irrigation, to high-precision technologies, such as computer-controlled systems*
- *Collaborate with medical or health researchers to address health problems related to geological materials or processes*
- *Develop ways to capture or use gases that are currently burned off as waste during oil production processes*
- *Participate in efforts to advance precision agriculture technology, such as developing advanced weed identification or automated spot spraying systems*
- *Program farm equipment, such as variable-rate planting equipment or pesticide sprayers, based on input from crop scouting and analysis of field condition variability*
- *Remove asbestos or lead from surfaces, using hand or power tools such as scrapers, vacuums, or high-pressure sprayers*
- *Operate machines or equipment to remove, package, store, or transport loads of waste materials*
- *Process e-waste, such as computer components containing lead or mercury*
- *Mix or pour concrete into forms to encase waste material for disposal*
- *Remove copper from circuit boards*

Table A7 – Selection of green skills with selected green specific tasks

| Item | Description | Beta | S.E. |
|---|---|---|---|
| 2B4g | Systems Analysis | 0.131*** | (0.0382) |
| 2B4h | Systems Evaluation | 0.135*** | (0.0405) |
| 2C10 | Transportation | 0.167** | (0.0647) |
| 2C3b | Engineering and Technology | 0.419*** | (0.111) |
| 2C3c | Design | 0.374*** | (0.0939) |
| 2C3d | Building and Construction | 0.442*** | (0.101) |
| 2C3e | Mechanical | 0.309*** | (0.111) |
| 2C4b | Physics | 0.409*** | (0.115) |
| 2C4d | Biology | 0.153*** | (0.0548) |
| 2C4g | Geography | 0.286*** | (0.0798) |
| 4A1b3 | Estimating the Quantifiable Characteristics of Products, Events, or Information | 0.110** | (0.0412) |
| 4A2a3 | Evaluating Information to Determine Compliance with Standards | 0.114*** | (0.0368) |
| 4A2b3 | Updating and Using Relevant Knowledge | 0.112** | (0.0422) |
| 4A2b4 | Developing Objectives and Strategies | 0.0981** | (0.0378) |
| 4A3b2 | Drafting, Laying Out, and Specifying Technical | 0.280*** | (0.0761) |

N=475 occupations (8-digit SOC). 3-digit SOC occupations with no green occupations are excluded. 3-digit SOC dummies included. OLS estimates. Standard errors clustered by 3-digit SOC in parenthesis. Beta and S.E. refer to the variable *Greenness*

Table A8 – Selection of green skills (with employment weights)

| Item | Description | Beta | S.E. |
|---|---|---|---|
| 2C3b | Engineering and Technology | 0.244*** | (0.0496) |
| 2C3c | Design | 0.206*** | (0.0638) |
| 2C3d | Building and Construction | 0.303*** | (0.0903) |
| 2C3e | Mechanical | 0.221*** | (0.0446) |
| 2C4b | Physics | 0.246*** | (0.0367) |
| 2C4c | Chemistry | 0.140*** | (0.0427) |
| 2C4d | Biology | 0.124*** | (0.0275) |
| 2C4g | Geography | 0.153*** | (0.0306) |

N=475 occupations (8-digit SOC). 3-digit SOC occupations with no green occupations are excluded. 3-digit SOC dummies included. OLS estimates weighted by employment share. Standard errors clustered by 3-digit SOC in parenthesis. Beta and S.E. refer to the variable *Greenness.*

Table A9 – Selection of green skills (count of specific tasks)

| Item | Description | Green specific tasks | | Total specific tasks | |
|---|---|---|---|---|---|
| | | Beta | S.E. | Beta | S.E. |
| 2B4h | Systems Evaluation | 0.00230** | (0.000840) | 0.00158** | (0.000716) |
| 2C3b | Engineering and Technology | 0.00836*** | (0.00240) | -0.000794 | (0.00119) |
| 2C3c | Design | 0.00718*** | (0.00202) | -0.000306 | (0.00150) |
| 2C3d | Building and Construction | 0.00931*** | (0.00221) | -0.00217 | (0.00128) |
| 2C3e | Mechanical | 0.00637** | (0.00233) | -0.00191 | (0.00124) |
| 2C4b | Physics | 0.00839*** | (0.00244) | -0.00134 | (0.000823) |
| 2C4g | Geography | 0.00681*** | (0.00146) | 0.000354 | (0.00107) |
| 2C8b | Law and Government | 0.00419*** | (0.00150) | 0.00102 | (0.00129) |
| 4A1b3 | Estimating the Quantifiable Characteristics of Products, Events, or Information | 0.00266** | (0.00103) | -0.000312 | (0.000760) |
| 4A2a3 | Evaluating Information to Determine Compliance with Standards | 0.00260*** | (0.000854) | 0.000859 | (0.000728) |
| 4A3a4 | Operating Vehicles, Mechanized Devices, or Equipment | 0.00520*** | (0.00149) | -0.000908 | (0.00124) |
| 4A3b2 | Drafting, Laying Out, and Specifying Technical Devices, Parts, and Equipment | 0.00570*** | (0.00163) | 0.0000792 | (0.00117) |
| 4A4b6 | Provide Consultation and Advice to Others | 0.00291*** | (0.000798) | 0.000844 | (0.00123) |

N=475 occupations (8-digit SOC). 3-digit SOC occupations with no green occupations are excluded. 3-digit SOC dummies included. OLS estimates weighted. Standard errors clustered by 3-digit SOC in parenthesis. Beta and S.E. refer to the variables *Count of green specific tasks* and *Count of total specific tasks*.

# Web Appendix B - Data

## *B1. O*NET and BLS data*

Our set of skill measures is built using occupation-MSA employment levels from BLS Occupational Employment Statistics (BLS-OES) for 2006-2014 to weight O*NET data on occupational skills. We use release 17.0 (July 2012) of O*NET.

Importance scores of selected skill measures range from 1 (not important) to 5 (very important) and measure how important the general task is for each occupation. Before computing $GGS_k$, we rescale these scores to range between 0 and 1 (we subtract 1 and divide by 4 each item that enters $GGS_k$). In addition to our *GGS* importance indices, we also build an index of Routine Task Intensity based on the items of O*NET identified by Acemoglu and Autor (2011). The list of items is reported in Table B1.

BLS-OES data at the metropolitan area level are released yearly since 1999. However, prior to 2006 metropolitan areas boundaries were defined differently and cannot be easily harmonized with the new delineation that has been adopted starting from 2006. Moreover, no information for non-metropolitan areas was available before 2006. Employment by occupation and metropolitan and non-metropolitan areas is reported if it is greater than 30 and if the cell occupation-area was 'sampled'. We filled missing values based on employment observed in the same cell occupation area in adjacent years or, if no information available, split employment in the cell macro-occupation (2-digit SOC) - area to 6-digit occupations based on federal-level employment shares. Finally, we employed a weighted crosswalk between SOC2009 and SOC2010 classification to obtain data in terms of SOC2010 occupations also for the period 2006-2009.

As noted in the main text, there is a mismatch between the aggregation of the O*NET database and the Occupational Employment Statistics. O*NET reports information for detailed 8-digit SOC occupations while the Occupational Employment Statistics provides employment data at the SOC 6-digit level only. To obtain area-level data, we must aggregate O*NET information at the 8-digit SOC to the 6-digit level to match employment statistics from BLS. This is clearly a limitation of the combination of O*NET with the BLS Occupational Employment Statistics Database.

In the analysis in the text, we assume that employment is uniformly distributed across all 8-digit occupations within the 6-digit category. As we discuss below, this approach may sometimes over-estimate the importance of (8-digit) green occupations within broader 6-digit occupations that also contain non-green occupations. Thus, in sensitivity analysis presented below, we instead assume that all employees work within the non-green 8-digit occupation(s) in these mixed 6-digit groups. Since green occupations are, by

construction, more green skills intensive, we consider this a lower bound of the importance of *GGS*. As we show below, our results are nearly identical using either assumption.

We begin by illustrating our assumptions with examples from the O*NET data. Note that there is no obvious solution to match the data, as the nature of 6-digit groups varies. In some cases, the different 8-digit occupations within the same 6-digit occupation represent distinct groups (as in examples 1 and 2 below). In other cases, the 8-digit occupation(s) represent a smaller sub-set of the larger 6-digit group (as in example #3 below). In each example below, occupations that make use of green skills are in italics.

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**Example 1:**

17-3023 Electrical and Electronic Engineering Technicians

17-3023.01 Electronics Engineering Technicians

*17-3023.03 Electrical Engineering Technicians*

In this example, O*NET does not provide skills information for the six-digit occupation "Electrical and Electronic Engineering Technicians". It only provides skills information for the two 8-digit subcategories. Of these two subcategories, only one is in our list of green occupations (in italics above). Since data on employment levels are only available at the 6-digit level, we must decide how to allocate employment between these two occupations. In the main text, we assume that employment in occupation 17-3023 is split equally, with one-half employed in category 17-3023.01 and one-half employed in category 17-3023.03. Thus, the importance of *Green General Skills* for the six-digit occupation 17-3023 is the average of the importance of *Green General Skills* in each of these two 8-digit categories. In the lower bound sensitivity analysis that follows, we assume all employment in this category goes to the non-green category 17-3023.01. Thus, the importance of *Green General Skills* for the six-digit occupation 17-3023 is the simply the importance of *Green General Skills* in 17-3023.01.

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**Example 2:**

| | |
|---|---|
| 11-3071 | Transportation, Storage, and Distribution Managers |
| *11-3071.01* | *Transportation Managers* |
| *11-3071.02* | *Storage and Distribution Managers* |
| *11-3071.03* | *Logistics Managers* |

Here again, O*NET does not provide skills information for the six-digit occupation "Transportation, Storage, and Distribution Managers". It only provides skills information for the three 8-digit subcategories. In this case, all three of these subcategories are occupations performing green specific tasks (i.e. green occupations). Thus, in both the main text and our lower bound analysis, we distribute one-third of employees in the six-digit category 11-3071 to each of the three 8-digit categories above. Thus, the importance of *Green General Skills* in the six-digit occupation 11-3071 is the simple average of the importance of these skills in the three eight-digit occupations above.

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**Example 3:**

19-3011 Economists

19-3011.00 Economists

*19-3011.01 Environmental Economists*

In contrast, in this example O*NET includes skills data for both the six-digit occupation "Economists" (classified as 19-3011.00) and for the eight-digit occupation "Environmental Economist". As we only have employment data at the six-digit level, we do not observe how many economists are environmental economists. Thus, we must decide how to allocate employment among environmental economists and all other economists. Our assumption in the text is clearly an upper bound in this case, as it allocates one-half of employment in this six-digit category to "Economists" and one-half to "Environmental Economists".[2] In contrast, our lower-bound sensitivity analysis assumes that no economists are environmental economists, instead allocating all employment to the broader 6-digit level of economists.

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While these assumptions mean that area employment data do not exactly match the level of detail in our skills data, note that this mismatch only affects a relatively small number of occupations. There is no 8-digit sub-category for 678 occupations (which employ 83.3 percent of US workers). For the remaining 97 6-digit occupations (employing the remaining 16.7 percent of US workers) the average number of 8-digit occupations is 3 and the median is 2, with a maximum of 12. While this requires us to make assumptions

[2] Note that this assumption need not always be an upper bound. For instance, in example 1 it would not be an upper bound if more workers are employed as Electrical Engineering Technicians (17-3023.03) than as Electronics Engineering Technicians (17-3023.01).

about the distribution of 8-digit occupations within these 6-digit occupation groups, the 6-digit occupation groups capture nearly all of the relevant variation in our skill measures.

To show this, we compute the standard deviation of the *GGS importance index* across 8-digit occupation groups and again across 6-digit occupational groups. The ratio of these two standard deviations provides a quantification of the heterogeneity in *GGS* importance across 8-digit occupations explained by the heterogeneity in *GGS* importance across 6-digit occupations. The share of variation in *GGS* importance explained by 6-digit occupations ranges from 97.2 to 98.8 percent.[3] If we repeat this exercise for the sub-set of 97 6-digit occupations with multiple 8-digit occupations (that only represent, however, only 16.7 percent of US workers), the 6-digit dimension still explains between 90.2 and 94.3 percent of the heterogeneity in *GGS* importance.[4]

Because skill variation at 6-digit level accounts for the bulk of variation at 8-digit level, the way 8-digit task intensity values are aggregated to the 6-digit level should have little influence on our results. As explained above, in our analysis, we calculate *GGS* importance at 6-digit SOC as the simple mean of the *GGS* importance at 8-digit SOC. The implicit assumption here is that employment is uniformly distributed across all 8-digit occupations within the 6-digit group. This approach, however, may sometimes over or under-estimate the importance of (8-digit) green occupations within broader 6-digit occupations that also contain non-green occupation.

We use the case of Electrical Engineering Technicians (SOC code: 17-3023.03) from Example #1 above to illustrate. This occupation has a *Greenness* score of 0.21. It belongs to a broader 6-digit SOC occupation that also includes the non-green occupation of Electronics Engineering Technicians (17-3023.01). By taking the simple average of *GGS* importance within the 6-digit SOC we are assuming that half of the employees in occupation 17-3023 are Electronics Engineering Technicians and half are Electrical Engineering Technicians. The intensity of three of our four green general skills is higher for the green occupation, Electrical Engineering Technicians, as shown in Table B2. Only for Engineering & Technical skills is the skill intensity higher for the non-green Electronics Engineers. However, the difference is small (0.48 vs 0.49). Thus, we cannot definitively state whether our assumption under-estimates or over-estimates the importance of each skill within the 6-diigt occupation, as this depends on whether electrical engineers truly represent more than half or less than half of employment in this category and on whether each skill is more or less important for electrical engineers.

[3] The exact figures are: 97.2 percent for Science, 98.1 percent for Monitoring, 98.5 percent for Engineering and Technical, and 98.8 percent for Operation and Management.

[4] Here, the exact figures are: 90.2 percent for Science, 91.3 percent for Monitoring, 94 percent for Engineering and Technical, and 94.3 percent for Operation and Management.

In summary, we believe that the choice of the method to calculate 6-digit task constructs for occupations with multiple 8-digit occupations should not influence our results because: i) differences in *GGS* importance across 6-digit occupations explain a very large share of the heterogeneity of *GGS* importance across 6-digit occupations, even when considering only 6-digit occupations with multiple 8-digit occupations; and ii) these occupations only represent a small part of US employment (16.7 percent overall, 2.9 percent in 6-digit occupations that contain both green and non-green 8-digit occupations).

Nonetheless, as a robustness check we re-run our analysis using *GGS* importance measures based upon the more conservative lower-bound assumption described above. Here, for those 6-digit occupations that contain both green and non-green 8-digit occupations, we make the extreme assumption that all employees work within the non-green 8-digit occupation(s). As green occupations are by construction more intensive in *GGS*, this should represent a lower bound of our *GGS* importance measures. Results based on these modified *GGS* importance are reported in Table B3. Results appear almost identical in magnitude and statistical significance to those of the baseline estimates of **Error! Reference source not found.**. This lends support to our claim that the results do not depend on the choice we made about the distribution of employees across 8-digit occupations within 6-digit occupations.

## *B2. Environmental Regulation*

Information on county-level nonattainment is retrieved from the 'Green Book Nonattainment Areas for Criteria Pollutants' maintained by the Environment Protection Agency (EPA) and available online at http://www3.epa.gov/airquality/greenbook/. Attainment status by county is extended to the whole metropolitan and non-metropolitan area sample as discussed in Section 3.1. Moreover, as discussed in Ferris et al. (2014), all counties and areas in the states included in the Ozone Transport Region have to implement regulatory actions equivalent to the ones mandated for nonattainment counties for the Ozone standards, even though they comply with the standard.

## *B3. Data sources for control variables*

Information on the distribution of employment by industry of metropolitan and non-metropolitan areas comes from the BLS Quarterly Census of Employment and Wages (CEW). We aggregated county-level figures to the metropolitan and non-metropolitan area level. Primary industries include NAICS codes 11 and 21, utilities NAICS codes 22 manufacturing industries NAICS codes 31-33. Also information on average establishment size (average employees per establishment) is retrieved from the BLS-CEW.

Data on resident population comes from the US Census Bureau. Once again, we aggregate county-level data to the metropolitan and non-metropolitan level.

Import penetration is measured as the ratio between import and 'domestic consumption' (import + domestic production - export) at the 4-digit NAICS level for year 2006. The data on total import and export for the US assembled by Schott (2008) are available at http://faculty.som.yale.edu/peterschott/sub_international.htm. Data on total production at the federal level by 4-digit NAICS manufacturing industries come from the NBER-CES database. We compute import penetration at the federal level and attribute it to metropolitan and non-metropolitan areas by multiplying industry-level import penetration by area-level employment share by 4-digit NAICS industry. This latter information, for year 2006, comes from the County Business Patterns database.

## *B5. Descriptive statistics for GGS importance*

We report some descriptive statistics about the distribution of *GGS* importance across metropolitan and non-metropolitan areas. In particular Table B4 shows the distribution of *GGS* importance (only for 'exposed' occupations), weighted by area employment, for 537 metropolitan and non-metropolitan areas and by year. Table B5 reports the cross-sectional correlation matrix across *GGS* importance (only for 'exposed' occupations) at the metropolitan and non-metropolitan area level weighted by area employment (average 2006-2014).

Table B1 - Items included in the Routine Task Intensity (RTI) index (source: Acemoglu and Autor, 2011)

| | |
|---|---|
| Non-routine analytical (NRA) | |
| 4A2a4 | Analyzing Data or Information |
| 4A2b2 | Thinking Creatively |
| 4A4a1 | Interpreting the Meaning of Information for Others |
| Non-routine interactive (NRI) | |
| 4A4a4 | Establishing and Maintaining Interpersonal Relationships |
| 4A4b4 | Guiding, Directing, and Motivating Subordinates |
| 4A4b5 | Coaching and Developing Others |
| Routine cognitive (RC) | |
| 4C3b4 (cx) | Importance of Being Exact or Accurate |
| 4C3b7 (cx) | Importance of Repeating Same Tasks |
| 4C3b8 (cx) | Structured versus Unstructured Work (reverse) |
| Routine manual (RM) | |
| 4A3a3 | Controlling Machines and Processes |
| 4C2d1i (cx) | Spend Time Making Repetitive Motions |
| 4C3d3 (cx) | Pace Determined by Speed of Equipment |

Table B2 – Example of multiple 8-digit occupations

| | Greenness | Engineering & Technical | Science | Operation Management | Monitoring |
|---|---|---|---|---|---|
| 17-3023.01 Electronics Engineering Tech. | 0 | 0.49 | 0.20 | 0.50 | 0.38 |
| 17-3023.03 Electrical Engineering Tech. | 0.21 | 0.48 | 0.24 | 0.53 | 0.45 |

Table B3 - Estimates for 'lower bound' *GGS* importance measures

| | Science | Engineering & technical | Operation management | Monitoring |
|---|---|---|---|---|
| NA in t=0 x trend | -0.000120 | -0.000210 | 0.0000112 | -0.0000237 |
| | (0.0000794) | (0.000144) | (0.000111) | (0.000115) |
| NA designation | 0.000363 | 0.00120* | 0.000351 | 0.000697* |
| | (0.000374) | (0.000682) | (0.000440) | (0.000381) |
| NA implementation | 0.000372 | 0.000936 | 0.000841* | 0.000274 |
| | (0.000386) | (0.000788) | (0.000459) | (0.000420) |
| NA designation + NA implementation | 0.000735** | 0.00213*** | 0.00119** | 0.000971** |
| | (0.000364) | (0.000657) | (0.000531) | (0.000465) |
| R sq | 0.332 | 0.6 | 0.615 | 0.541 |
| N | 4824 | 4806 | 4797 | 4824 |

Dependent variables refer to SOC codes (2-digit) with at least one green job (SOC codes: 11, 13, 15, 17, 19, 23, 27, 41, 43, 47, 49, 51, 53). *GGS* measures calculated by setting to zero the employment of 8-digit green occupations that belong to 6-digit occupations with at least one non-green occupation. Fixed effect model weighted by kernel-based weights based on propensity score. Other control variables: state-specific year dummies; other controls interacted with linear trend: share of manufacturing (2005), share of primary sector (2005), share of construction sector (2005), share of utility sector (2005), import penetration (2005), log of population density (2005), log of average establishment size (2005), exposure to the crisis (2005).

Table B4 – Descriptive statistics of *GGS* importance by metropolitan and non-metropolitan area

| Year | Min | Q1 | Median | Q3 | Max | Average | SD | IQR |
|---|---|---|---|---|---|---|---|---|
| Engineering and Technical | | | | | | | | |
| 2006 | 0.247 | 0.271 | 0.279 | 0.287 | 0.443 | 0.279 | 0.014 | 0.015 |
| 2007 | 0.251 | 0.272 | 0.280 | 0.288 | 0.353 | 0.280 | 0.014 | 0.016 |
| 2008 | 0.235 | 0.271 | 0.278 | 0.287 | 0.361 | 0.279 | 0.015 | 0.016 |
| 2009 | 0.248 | 0.268 | 0.277 | 0.285 | 0.366 | 0.278 | 0.014 | 0.017 |
| 2010 | 0.242 | 0.265 | 0.273 | 0.283 | 0.365 | 0.275 | 0.014 | 0.018 |
| 2011 | 0.241 | 0.265 | 0.274 | 0.283 | 0.361 | 0.275 | 0.015 | 0.019 |
| 2012 | 0.241 | 0.265 | 0.275 | 0.284 | 0.361 | 0.276 | 0.015 | 0.019 |
| 2013 | 0.240 | 0.266 | 0.276 | 0.283 | 0.412 | 0.277 | 0.015 | 0.018 |
| 2014 | 0.234 | 0.267 | 0.277 | 0.285 | 0.409 | 0.278 | 0.015 | 0.018 |
| Total | 0.234 | 0.268 | 0.277 | 0.285 | 0.443 | 0.277 | 0.015 | 0.017 |
| Science | | | | | | | | |
| 2006 | 0.100 | 0.110 | 0.113 | 0.119 | 0.170 | 0.114 | 0.008 | 0.008 |
| 2007 | 0.101 | 0.111 | 0.113 | 0.119 | 0.165 | 0.115 | 0.008 | 0.009 |
| 2008 | 0.090 | 0.111 | 0.114 | 0.120 | 0.166 | 0.115 | 0.009 | 0.009 |
| 2009 | 0.097 | 0.110 | 0.114 | 0.120 | 0.172 | 0.116 | 0.009 | 0.010 |
| 2010 | 0.098 | 0.109 | 0.113 | 0.119 | 0.171 | 0.115 | 0.009 | 0.010 |
| 2011 | 0.093 | 0.109 | 0.113 | 0.120 | 0.170 | 0.115 | 0.009 | 0.011 |
| 2012 | 0.091 | 0.110 | 0.114 | 0.120 | 0.176 | 0.115 | 0.009 | 0.011 |
| 2013 | 0.099 | 0.110 | 0.114 | 0.121 | 0.207 | 0.116 | 0.009 | 0.011 |
| 2014 | 0.098 | 0.111 | 0.114 | 0.122 | 0.206 | 0.116 | 0.009 | 0.011 |
| Total | 0.090 | 0.110 | 0.114 | 0.120 | 0.207 | 0.115 | 0.009 | 0.010 |
| Operation management | | | | | | | | |
| 2006 | 0.379 | 0.423 | 0.435 | 0.446 | 0.507 | 0.435 | 0.018 | 0.023 |
| 2007 | 0.380 | 0.424 | 0.437 | 0.448 | 0.503 | 0.436 | 0.018 | 0.024 |
| 2008 | 0.381 | 0.426 | 0.438 | 0.450 | 0.504 | 0.438 | 0.018 | 0.024 |
| 2009 | 0.386 | 0.428 | 0.442 | 0.453 | 0.513 | 0.441 | 0.019 | 0.025 |
| 2010 | 0.385 | 0.429 | 0.444 | 0.453 | 0.522 | 0.442 | 0.020 | 0.024 |
| 2011 | 0.381 | 0.430 | 0.445 | 0.454 | 0.525 | 0.443 | 0.020 | 0.025 |
| 2012 | 0.381 | 0.430 | 0.446 | 0.455 | 0.524 | 0.444 | 0.020 | 0.025 |
| 2013 | 0.385 | 0.430 | 0.447 | 0.455 | 0.528 | 0.444 | 0.020 | 0.025 |
| 2014 | 0.390 | 0.431 | 0.447 | 0.456 | 0.525 | 0.445 | 0.021 | 0.025 |
| Total | 0.379 | 0.428 | 0.442 | 0.452 | 0.528 | 0.441 | 0.020 | 0.024 |
| Monitoring | | | | | | | | |
| 2006 | 0.394 | 0.435 | 0.445 | 0.451 | 0.483 | 0.443 | 0.013 | 0.016 |
| 2007 | 0.393 | 0.436 | 0.446 | 0.452 | 0.486 | 0.444 | 0.013 | 0.016 |
| 2008 | 0.397 | 0.438 | 0.445 | 0.452 | 0.484 | 0.444 | 0.013 | 0.014 |
| 2009 | 0.397 | 0.440 | 0.447 | 0.455 | 0.488 | 0.447 | 0.014 | 0.015 |
| 2010 | 0.396 | 0.441 | 0.448 | 0.456 | 0.492 | 0.448 | 0.014 | 0.015 |
| 2011 | 0.399 | 0.441 | 0.449 | 0.457 | 0.503 | 0.448 | 0.014 | 0.016 |
| 2012 | 0.398 | 0.441 | 0.451 | 0.458 | 0.503 | 0.449 | 0.014 | 0.016 |
| 2013 | 0.401 | 0.441 | 0.451 | 0.457 | 0.494 | 0.449 | 0.014 | 0.016 |
| 2014 | 0.402 | 0.441 | 0.451 | 0.458 | 0.493 | 0.450 | 0.014 | 0.016 |
| Total | 0.393 | 0.439 | 0.448 | 0.456 | 0.503 | 0.447 | 0.014 | 0.016 |

Statistics weighted by total employment. N=537 metropolitan and nonmetropolitan areas. Skill measures for occupations in SOC codes (2-digit) with at least one green job (SOC codes: 11, 13, 15, 17, 19, 23, 27, 41, 43, 47, 49, 51, 53).

Table B5 – Correlation between *GGS* importance measures at the metropolitan and non-metropolitan area level

| | Engineering & Technical | Science | Operation management | Monitoring |
|---|---|---|---|---|
| Engineering & Technical | 1.000 | | | |
| Science | 0.8807 | 1.000 | | |
| Operation management | -0.0951 | 0.1284 | 1.000 | |
| Monitoring | -0.1695 | 0.0943 | 0.8988 | 1.000 |

N=537. Correlation on average 2006-2014 values weighted by average employment. Skill measures for occupations in SOC codes (2-digit) with at least one green job (SOC codes: 11, 13, 15, 17, 19, 23, 27, 41, 43, 47, 49, 51, 53).

# Web Appendix C - Brown jobs

As discussed in Section 2.5, we label occupations that are specific to pollution-intensive industries as 'brown'. The first step is to identify pollution intensive industries. To this end, we retrieve facility-level information on air emissions from the National Emission Inventory database (year 2011) and the Greenhouse Gas Emissions from Large Facilities (year 2011), both maintained by the Environmental Protection Agency and aggregate up emissions at the national level to the main NAICS code (4-digit) of the facility. We focus on substances that affect the concentration of the 6 criteria pollutants (CO, VOC, NOx, SO2, PM10, PM2.5 and lead) and CO2 emissions. For each one of them, we compute emission intensity (per employee, source: BLS) and label 'pollution intensive' the industries for which emission intensity lied in the top 5 percent of the (weighted) distribution. Finally, we define 'brown industries' those for which at least three substances were in the top 5 percent. The list of brown industries, together with the substances for which the industry is in the top 5 percent, is reported in Table C1.

As a second step, we calculated for each 6-digit SOC occupation the share of total employees of occupations that are employed in any of the brown industries using the BLS-OES estimates of occupational employment by 4-digit NAICS industries for years 2006-2014. We identify brown jobs by selecting occupations with relative share of employees in brown industries at least 7 times larger than the share of employees in brown industries for all occupations. The list of 6-digit SOC brown occupations is reported in Table C2.

Table C1 - Brown industries

| NAICS | Description | Count of pollutants above 95th percentile | List of pollutants above 95th percentile |
|---|---|---|---|
| 2122 | Metal Ore Mining | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 2123 | Nonmetallic Mineral Mining and Quarrying | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 2211 | Electric Power Generation, Transmission and Distribution | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 2213 | Water, Sewage and Other Systems | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3112 | Grain and Oilseed Milling | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3113 | Sugar and Confectionery Product Manuf | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3114 | Fruit and Vegetable Preserving and Specialty Food Manuf | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3121 | Beverage Manuf | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3133 | Textile and Fabric Finishing and Fabric Coating Mills | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3212 | Veneer, Plywood, and Engineered Wood Product Manuf | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3221 | Pulp, Paper, and Paperboard Mills | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3241 | Petroleum and Coal Products Manuf | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3251 | Basic Chemical Manuf | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3252 | Resin, Synthetic Rubber, and Artificial Synthetic Fibers and Filaments Manuf | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3253 | Pesticide, Fertilizer, and Other Agricultural Chemical Manuf | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3259 | Other Chemical Product and Preparation Manuf | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3271 | Clay Product and Refractory Manuf | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3272 | Glass and Glass Product Manuf | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3273 | Cement and Concrete Product Manuf | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3274 | Lime and Gypsum Product Manuf | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3279 | Other Nonmetallic Mineral Product Manuf | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3311 | Iron and Steel Mills and Ferroalloy Manuf | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3312 | Steel Product Manuf from Purchased Steel | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3313 | Alumina and Aluminum Production and Processing | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3314 | Nonferrous Metal (except Aluminum) Production and Processing | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 3315 | Foundries | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 5622 | Waste Treatment and Disposal | 8 | CO VOC NOX SO2 PM10 PM2.5 Lead CO2 |
| 2111 | Oil and Gas Extraction | 7 | CO VOC NOX SO2 PM10 PM2.5 CO2 |
| 2212 | Natural Gas Distribution | 7 | CO VOC NOX SO2 PM10 PM2.5 CO2 |
| 3119 | Other Food Manuf | 7 | CO NOX SO2 PM10 PM2.5 Lead CO2 |
| 3122 | Tobacco Manuf | 7 | CO VOC NOX SO2 PM10 PM2.5 CO2 |
| 3222 | Converted Paper Product Manuf | 7 | CO VOC NOX PM10 PM2.5 Lead CO2 |
| 3256 | Soap, Cleaning Compound, and Toilet Preparation Manuf | 7 | CO NOX SO2 PM10 PM2.5 Lead CO2 |
| 3262 | Rubber Product Manuf | 7 | CO VOC NOX PM10 PM2.5 Lead CO2 |
| 3361 | Motor Vehicle Manuf | 7 | CO VOC NOX PM10 PM2.5 Lead CO2 |
| 3365 | Railroad Rolling Stock Manuf | 7 | CO VOC NOX PM10 PM2.5 Lead CO2 |
| 4862 | Pipeline Transportation of Natural Gas | 7 | CO VOC NOX SO2 PM10 PM2.5 CO2 |
| 4881 | Support Activities for Air Transportation | 7 | CO VOC NOX SO2 PM10 PM2.5 Lead |
| 4882 | Support Activities for Rail Transportation | 7 | CO VOC NOX SO2 PM10 PM2.5 Lead |
| 2121 | Coal Mining | 6 | CO NOX SO2 PM10 PM2.5 CO2 |
| 3161 | Leather and Hide Tanning and Finishing | 6 | CO VOC NOX PM10 PM2.5 CO2 |
| 3211 | Sawmills and Wood Preservation | 6 | CO VOC NOX PM10 PM2.5 Lead |
| 3219 | Other Wood Product Manuf | 6 | CO VOC NOX PM10 PM2.5 Lead |
| 3328 | Coating, Engraving, Heat Treating, and Allied Activities | 6 | CO VOC NOX PM10 PM2.5 Lead |
| 3359 | Other Electrical Equipment and Component Manuf | 6 | CO NOX SO2 PM10 PM2.5 Lead |
| 4861 | Pipeline Transportation of Crude Oil | 6 | CO VOC NOX PM10 PM2.5 CO2 |
| 4869 | Other Pipeline Transportation | 6 | CO VOC NOX PM10 PM2.5 CO2 |
| 3111 | Animal Food Manuf | 5 | CO NOX PM10 PM2.5 CO2 |
| 3115 | Dairy Product Manuf | 5 | CO NOX PM10 PM2.5 CO2 |
| 3132 | Fabric Mills | 5 | CO NOX PM10 PM2.5 CO2 |
| 3141 | Textile Furnishings Mills | 5 | NOX SO2 PM10 Lead CO2 |
| 3254 | Pharmaceutical and Medicine Manuf | 5 | CO NOX SO2 PM10 CO2 |
| 3369 | Other Transportation Equipment Manuf | 5 | CO VOC PM10 PM2.5 Lead |
| 5629 | Remediation and Other Waste Management Services | 5 | CO NOX PM10 PM2.5 Lead |
| 3255 | Paint, Coating, and Adhesive Manuf | 4 | VOC PM10 PM2.5 Lead |
| 3329 | Other Fabricated Metal Product Manuf | 4 | CO PM10 PM2.5 Lead |
| 4247 | Petroleum and Petroleum Products Merchant Wholesalers | 4 | CO VOC NOX CO2 |
| 2131 | Support Activities for Mining | 3 | CO NOX CO2 |
| 3321 | Forging and Stamping | 3 | CO NOX CO2 |
| 3322 | Cutlery and Handtool Manuf | 3 | CO PM10 PM2.5 |
| 3366 | Ship and Boat Building | 3 | VOC PM10 PM2.5 |
| 3371 | Household and Institutional Furniture and Kitchen Cabinet Manuf | 3 | VOC PM10 PM2.5 |

Table C2 - Brown occupations

| SOC 2010 | Title |
|---|---|
| 17-2041 | Chemical Engineers |
| 17-2151 | Mining and Geological Engineers, Including Mining Safety Engineers |
| 17-2171 | Petroleum Engineers |
| 19-1012 | Food Scientists and Technologists |
| 19-2031 | Chemists |
| 19-4031 | Chemical Technicians |
| 43-5041 | Meter Readers, Utilities |
| 45-4023 | Log Graders and Scalers |
| 47-4071 | Septic Tank Servicers and Sewer Pipe Cleaners |
| 47-5011 | Derrick Operators, Oil and Gas |
| 47-5012 | Rotary Drill Operators, Oil and Gas |
| 47-5013 | Service Unit Operators, Oil, Gas, and Mining |
| 47-5021 | Earth Drillers, Except Oil and Gas |
| 47-5031 | Explosives Workers, Ordnance Handling Experts, and Blasters |
| 47-5042 | Mine Cutting and Channeling Machine Operators |
| 47-5051 | Rock Splitters, Quarry |
| 47-5061 | Roof Bolters, Mining |
| 47-5071 | Roustabouts, Oil and Gas |
| 47-5081 | Helpers--Extraction Workers |
| 49-2095 | Electrical and Electronics Repairers, Powerhouse, Substation, and Relay |
| 49-9012 | Control and Valve Installers and Repairers, Except Mechanical Door |
| 49-9041 | Industrial Machinery Mechanics |
| 49-9043 | Maintenance Workers, Machinery |
| 49-9045 | Refractory Materials Repairers, Except Brickmasons |
| 49-9051 | Electrical Power-Line Installers and Repairers |
| 49-9093 | Fabric Menders, Except Garment |
| 51-1011 | First-Line Supervisors of Production and Operating Workers |
| 51-2091 | Fiberglass Laminators and Fabricators |
| 51-3091 | Food and Tobacco Roasting, Baking, and Drying Machine Operators and Tenders |
| 51-3092 | Food Batchmakers |
| 51-3093 | Food Cooking Machine Operators and Tenders |
| 51-4021 | Extruding and Drawing Machine Setters, Operators, and Tenders, Metal and Plastic |
| 51-4022 | Forging Machine Setters, Operators, and Tenders, Metal and Plastic |
| 51-4023 | Rolling Machine Setters, Operators, and Tenders, Metal and Plastic |
| 51-4033 | Grinding, Lapping, Polishing, and Buffing Machine Tool Setters, Operators, and Tenders, Metal and Plastic |
| 51-4051 | Metal-Refining Furnace Operators and Tenders |
| 51-4052 | Pourers and Casters, Metal |
| 51-4062 | Patternmakers, Metal and Plastic |
| 51-4071 | Foundry Mold and Coremakers |
| 51-4191 | Heat Treating Equipment Setters, Operators, and Tenders, Metal and Plastic |
| 51-4192 | Layout Workers, Metal and Plastic |
| 51-4193 | Plating and Coating Machine Setters, Operators, and Tenders, Metal and Plastic |
| 51-4194 | Tool Grinders, Filers, and Sharpeners |
| 51-6061 | Textile Bleaching and Dyeing Machine Operators and Tenders |
| 51-6063 | Textile Knitting and Weaving Machine Setters, Operators, and Tenders |
| 51-6064 | Textile Winding, Twisting, and Drawing Out Machine Setters, Operators, and Tenders |
| 51-6091 | Extruding and Forming Machine Setters, Operators, and Tenders, Synthetic and Glass Fibers |
| 51-6093 | Upholsterers |
| 51-7011 | Cabinetmakers and Bench Carpenters |
| 51-7021 | Furniture Finishers |
| 51-7031 | Model Makers, Wood |
| 51-7032 | Patternmakers, Wood |
| 51-7041 | Sawing Machine Setters, Operators, and Tenders, Wood |
| 51-7042 | Woodworking Machine Setters, Operators, and Tenders, Except Sawing |
| 51-8012 | Power Distributors and Dispatchers |
| 51-8091 | Chemical Plant and System Operators |
| 51-8092 | Gas Plant Operators |
| 51-8093 | Petroleum Pump System Operators, Refinery Operators, and Gaugers |
| 51-9011 | Chemical Equipment Operators and Tenders |
| 51-9012 | Separating, Filtering, Clarifying, Precipitating, and Still Machine Setters, Operators, and Tenders |
| 51-9021 | Crushing, Grinding, and Polishing Machine Setters, Operators, and Tenders |
| 51-9022 | Grinding and Polishing Workers, Hand |
| 51-9023 | Mixing and Blending Machine Setters, Operators, and Tenders |
| 51-9031 | Cutters and Trimmers, Hand |
| 51-9032 | Cutting and Slicing Machine Setters, Operators, and Tenders |
| 51-9041 | Extruding, Forming, Pressing, and Compacting Machine Setters, Operators, and Tenders |
| 51-9051 | Furnace, Kiln, Oven, Drier, and Kettle Operators and Tenders |

| SOC 2010 | Title |
|---|---|
| 51-9111 | Packaging and Filling Machine Operators and Tenders |
| 51-9121 | Coating, Painting, and Spraying Machine Setters, Operators, and Tenders |
| 51-9191 | Adhesive Bonding Machine Operators and Tenders |
| 51-9192 | Cleaning, Washing, and Metal Pickling Equipment Operators and Tenders |
| 51-9193 | Cooling and Freezing Equipment Operators and Tenders |
| 51-9195 | Molders, Shapers, and Casters, Except Metal and Plastic |
| 51-9196 | Paper Goods Machine Setters, Operators, and Tenders |
| 51-9197 | Tire Builders |
| 53-4013 | Rail Yard Engineers, Dinkey Operators, and Hostlers |
| 53-7031 | Dredge Operators |
| 53-7032 | Excavating and Loading Machine and Dragline Operators |
| 53-7033 | Loading Machine Operators, Underground Mining |
| 53-7041 | Hoist and Winch Operators |
| 53-7063 | Machine Feeders and Offbearers |
| 53-7071 | Gas Compressor and Gas Pumping Station Operators |
| 53-7072 | Pump Operators, Except Wellhead Pumpers |
| 53-7073 | Wellhead Pumpers |
| 53-7111 | Mine Shuttle Car Operators |
| 17-2041 | Chemical Engineers |
| 17-2151 | Mining and Geological Engineers, Including Mining Safety Engineers |

Table C3 - Detailed occupations used for 3-digit comparison

| SOC 3-digit | Green | Brown |
|---|---|---|
| 17-2 | • Aerospace Engineers<br>• Civil Engineers<br>• Electrical Engineers<br>• Electronics Engineers, Except Computer<br>• Environmental Engineers<br>• Mechanical Engineers<br>• Nuclear Engineers<br>• Engineers, All Other | • Chemical Engineers<br>• Mining and Geological Engineers, Including Mining Safety Engineers<br>• Petroleum Engineers |
| 19-1 | • Soil and Plant Scientists<br>• Conservation Scientists | • Food Scientists and Technologists |
| 19-2 | • Atmospheric and Space Scientists<br>• Environmental Scientists and Specialists, Including Health<br>• Geoscientists, Except Hydrologists and Geographers | • Chemists |
| 19-4 | • Nuclear Technicians<br>• Environmental Science and Protection Technicians, Including Health<br>• Life, Physical, and Social Science Technicians, All Other | • Chemical Technicians<br>• Geological and Petroleum Technicians |

# Web Appendix D - Results for propensity score matching

Table D1 - Estimates of *GGS*-specific propensity scores

| | (1) Engineering | (2) Science | (3) Op.&Manag. | (4) Monitoring |
|---|---|---|---|---|
| log(pop density) | 0.143* | 0.151** | 0.0843 | 0.131* |
| | (0.0732) | (0.0747) | (0.0737) | (0.0713) |
| | [balanced] | [balanced] | [balanced] | [balanced] |
| Share manuf sect | 1.483 | 1.783 | 4.181*** | 5.170*** |
| | (1.221) | (1.160) | (1.438) | (1.614) |
| | [balanced] | [balanced] | [balanced] | [balanced] |
| Share primary sect | -2.466 | -2.522 | -1.321 | -2.108 |
| | (2.935) | (2.952) | (2.925) | (3.024) |
| | [balanced] | [balanced] | [balanced] | [balanced] |
| Share utility sect | 47.94** | 47.63** | 54.31*** | 53.04*** |
| | (18.66) | (18.64) | (18.73) | (18.77) |
| | [balanced] | [balanced] | [balanced] | [balanced] |
| Share construction sect | 2.984 | 3.381 | 5.033 | 4.245 |
| | (4.283) | (4.112) | (4.161) | (4.097) |
| | [balanced] | [balanced] | [balanced] | [balanced] |
| log(estab size) | -0.0622** | -0.0623** | -0.0738** | -0.0778*** |
| | (0.0283) | (0.0283) | (0.0289) | (0.0291) |
| | [balanced] | [balanced] | [balanced] | [balanced] |
| Import penetration | -5.637 | -5.521 | -5.356 | -5.684 |
| | (4.066) | (4.068) | (4.071) | (4.070) |
| | [balanced] | [balanced] | [balanced] | [balanced] |
| Area is NA in 2006 | 0.504*** | 0.504*** | 0.479*** | 0.491*** |
| | (0.162) | (0.162) | (0.164) | (0.164) |
| | [balanced] | [balanced] | [balanced] | [balanced] |
| Engineering & technical | 2.850 | | | |
| | (4.577) | | | |
| | [balanced] | | | |
| Science | | 6.022 | | |
| | | (8.030) | | |
| | | [balanced] | | |
| Operation management | | | 16.57*** | |
| | | | (5.625) | |
| | | | [balanced] | |
| Monitoring | | | | 24.40*** |
| | | | | (7.835) |
| | | | | [balanced] |
| Pseudo R sq | 0.0877 | 0.0881 | 0.108 | 0.111 |
| N | 537 | 537 | 537 | 537 |

Probit model for year 2006. Standard errors in parenthesis. Post-matching balancing of covariates in brakets. * p<0.1, ** p<0.05, *** p<0.01. Number of observations: 537. Matching on propensity score based on kernel.

# Web Appendix E - Additional results

Our results depend on several assumptions that are necessary to deal with the data limitations mentioned above. We explore the robustness of our results to three of these assumptions: (1) our groupings of *Green General Skills*, (2) occupations exposed to environmental regulation, (3) our treatment of areas already designated as nonattainment areas prior to changes in the air quality standards, and (4) the timing of the effect of regulation. We briefly summarize here our main results.

Considering the effect of nonattainment designation on different skill items of our four *GGS* allows to highlight two policy relevant implications. First, the effect of nonattainment status on high-level design, engineering and technology skills is greater than on low-skilled engineering and technology skills. An implication of this is that public policy in support of the green economy should improve accessibility to advanced educational programs in engineering, design and architecture. Second, we observe a fair amount of heterogeneity within the Science and Monitoring groups. Within Science, physics is solely responsible for the positive effect of regulation, plausibly due to our focus on air pollution. One might expect biology to play a more important role responding to water pollution regulation, although the role of basic science may be limited in the context of short-term responses to policy shocks. Within Monitoring, nonattainment status increases the importance of compliance skills but not of legal skills. Our explanation for these findings is that compliance is bound to a specific site whereas the legal activities associated with it can be carried out remotely, such as in state capitals or company headquarters.

As a second robustness check we further tighten our definition of exposed occupations by just considering those occupations that belong to a 3-digit SOC group containing at least one green occupation, rather than a 2-digit SOC group containing a green occupation. Our main results remain qualitatively unaffected, with regulation having a positive impact on the demand of *GGS* within narrowly defined occupational groups. The effect on Science, however, is no longer statistically significant and, overall, the relative changes are slightly smaller when using a narrower set of occupations than the ones estimated in our baseline specification.

Third, we test the validity of our findings for different definitions of nonattainment designation. In our main analysis, we include in the treated group only areas that switch from attainment to nonattainment under one of the new standards. If we include in the treatment group areas that were already in nonattainment status under the old NAAQS standards, results remain qualitatively similar but we observe a reduction in the size of the effects of nonattainment designation on workforce composition. A possible explanation is that plants previously exposed to more stringent NAAQS regulation are likely to have made prior adjustments to their input mix to cope with environmental constraints and opportunities.

Finally, we re-estimate equation (4) adding a general interaction between the nonattainment designation and time dummies. In general, the effect of environmental regulation is strongest in the first four years after the policy change. Also encouraging is that environmental regulation has no effect on the importance of *GGS* prior to the policy shock, suggesting that our identification strategy does control for potential endogeneity in the setting of environmental regulations, as we see no evidence of different pre-treatment trends in treated and control areas.

Overall, this set of extensions confirms that the importance of Engineering and Technical skills increases due to environmental regulation. The skill gaps induced by environmental regulation on the other three *GGS* importance indices are smaller and less stable across specifications. There is thus insufficient evidence to recommend an expansion of training in these domains. We discuss these analyses in further detail below.

*Heterogeneous effects for specific skills.* Our groupings of *Green General Skills* bring together heterogeneous skill items on which environmental regulation may have different effects. For example, our first Green General Skill, Engineering and Technical skills, contains two skill items, 'Engineering and Technology' and 'Design', that have a significantly higher educational requirement than the remaining items. Clearly, the policy implications of our study for the adjustment in the educational supply would differ substantially depending on whether the increase in the demand of engineering skills can be met through expanding vocational and technical education rather than the set-up of academic programs. Similarly, our Monitoring group combines legal skills with compliance skills that are arguably more technical and require less formal training.

We re-estimate equation (4) for specific skill items to test for heterogeneous effects. Propensity score weights are built including the level of each specific item before the shock to estimate these weights. For ease of comparison, Table E1 presents the results quantified in terms of percentile rank changes, using the same methodology as in **Error! Reference source not found.**. Skills in italics denote that the sum of the designation and the implementation dummy effects are statistically significant. We highlight two important results. First, the effect of nonattainment status on high-level engineering and technology skills is greater than on low-skilled engineering and technology skills. We observe over a 5 percentile change in importance for 'Design' and 'Engineering', compared to 2 percentiles change in importance for 'Mechanical' and a 4 percentile change for 'Drafting'. An implication of this is that public policy in support of the green economy should improve accessibility to advanced educational programs in engineering, design and architecture. Second, we observe a fair amount of heterogeneity within the Science and Monitoring groups. Within Science, physics is solely responsible for the positive effect of regulation, plausibly due to our focus on air pollution. One might expect biology to play a more important role responding to water pollution regulation, although the role of basic science may be limited in the context of short-term responses to policy shocks.

Within Monitoring, nonattainment status increases the importance of compliance skills but not of legal skills. Our explanation for these findings is that compliance is bound to a specific site whereas the legal activities associated with it can be carried out remotely, such as in state capitals or company headquarters. Notably, the effect of nonattainment status on the specific skill of compliance is more than twice that of Monitoring in **Error! Reference source not found.** and close to that of our top Engineering and Technical skill items. Finally, note that the effect of regulation on the two skill not selected using principal component analysis, Geography and Operating Vehicles and Mechanized Devices, is small and statistically insignificant.

*Green vs. brown jobs*. An issue of our empirical strategy is that we cannot precisely identify those jobs directly or indirectly exposed to regulation. To overcome this shortcoming, we focus on potentially exposed occupations. Our *GGS* constructs for each metro and non-metro area include only 2-digit occupational groups containing at least one green job. Since there is little existing empirical work to identify occupations that are green or brown, the validity of our results may depend on this assumption.

Here we test the robustness of our main results using a different definition of exposed occupations. In particular, we build each area's *GGS importance index* using only 2-digit occupations that contain a brown job as defined in section 2.4. These include SOC-17 Engineering, SOC-19 Science, SOC-47 Construction, SOC-49 Repair and Maintenance, SOC-51 Production, SOC-53 Transport. Brown jobs should be the first jobs to be curtailed as a consequence of environmental regulation. Although a worker who has been displaced under these circumstances can in principle be re-employed elsewhere, labor research suggests that worker skills are strongly occupation-specific and portable only to jobs with a similar skill requirement (Gathmann and Schönberg, 2010). Limiting our analysis to 2-digit groups with at least one brown jobs accounts for these occupation-specific skill barriers to worker relocation. At the same time, this choice likely provides a very conservative estimate of the effect of environmental regulation on the demand of *GGS* because it overlooks potential skill gaps due to the creation of new green jobs in business services. In that sector, management, sales and legal services are relevant green occupations that do not have a corresponding brown job at 2-digit level. Thus, this strategy captures the ease of mobility for workers negatively affected by environmental regulation but ignores job creation in sectors unrelated to those workers that have been potentially laid off.

Table E4 presents the main results of this exercise. We again focus our comments on the quantification of the effect in terms of percentile change from the initial distribution. Narrowing the group of occupations considered in our comparison substantially reduces the size of the estimated effects. The effect of environmental regulation is now only significant for Engineering and Technical skills but the magnitude of the effect is smaller, with a percentile change of 2.9 points rather than 4.7 points. This resonates with our

descriptive evidence showing that, within narrowly defined occupational groups, the skill difference between brown and green jobs is small. Interestingly, the insignificant effect on Monitoring skills conceals a very large (an increase of 8.2 percentiles) and significant effect on Compliance skills (see last row of Table E4).

In addition, as a second robustness check we further tighten our definition of exposed occupations by just considering those occupations that belong to a 3-digit SOC group containing at least one green occupation, rather than a 2-digit SOC group containing a green occupation. In our baseline estimates, exposed occupations accounted for about 66 percent of US workers in 2006-2014, while selecting 3-digit SOC groups with at least one green occupation leads to include about 38 percent of US workers over the same period. Results are reported in Table E5.

Our main results remain qualitatively unaffected, with regulation having a positive impact on the demand of *GGS* within narrowly defined occupational groups. The effect on Science, however, is no longer statistically significant and, overall, the relative changes with respect to the initial *GGS* importance distribution predicted by environmental regulation are slightly smaller when using a narrower set of occupations than the ones estimated in our baseline specification.

*Different definitions of nonattainment designation.* Finally, we test the validity of our findings for different definitions of nonattainment designation. In our main analysis, we include in the treated group only areas that switch from attainment to nonattainment under one of the new standards. This assumption excludes from the treatment group areas that were already in nonattainment status under the old NAAQS standards. Here we include in the treated group all areas designated as nonattainment under one of the new standards. Table E6 summarizes the results again in terms of percentile changes in the initial ranking of regions. Note first that the overall effect on the treated is now the result of the effects on both new nonattainment areas (67) and nonattainment areas that experienced an increase in the stringency of NAAQS regulation (113). Across the board, we observe a reduction in the size of the effects of nonattainment designation on workforce composition. A possible explanation is that plants previously exposed to more stringent NAAQS regulation are likely to have made prior adjustments to their input mix to cope with environmental constraints and opportunities. Only for Science are the results quantitatively similar, with an increase of 2.7 percentage points here, compared to 2.8 before. The only qualitative change is that environmental regulation now has no significant effect on Monitoring skills.

To further explore the variation of effects over time, we re-estimate equation (4) adding a general interaction between the nonattainment designation and time dummies.[5] This analysis is only indicative of the effects of time as overlapping policy changes for different pollutants do not allow us to completely distinguish between the time profile of the policy effects and the heterogeneous effects across pollutants.

Table E7 presents the results using our main definition of nonattainment designation. In general, the effect of environmental regulation is strongest in the first four years after the policy change, especially for Compliance. Except for Engineering and Technical skills, the magnitude of the effect of regulation decreases beyond four years. Also encouraging is that environmental regulation has no effect on the *GGS importance index* prior to the policy shock. Thus, our main finding here is that our identification strategy does control for potential endogeneity in the setting of environmental regulations, as we see no evidence of different pre-treatment trends in treated and control areas. At the same time, the short time span of our analysis and the overlapping of different shocks do not allow us to give conclusive policy advice on the opportunity to start large scale educational programs for the need of the green economy. This open issue is left for future research.

[5] The new estimation equation is:

$$y_{jt} = \sum_{t \geq t_{NA}} \beta^t \text{NA_designation}_{j,t \geq t_{NA}} + \sum_{t \geq t_{impl}} \phi^t \text{NA_designation}_{j,t \geq t_{NA}} \times \text{NA_implement} +$$

$$+ \varphi \text{NA}_{j0} trend_t + \boldsymbol{\gamma} \mathbf{X}_{\text{j0}}\, trend_t + \mu_j + \mu_{ts} + \varepsilon_{jt}$$

Table E1 - Quantification of the effect of environmental regulation on *GGS* importance (by item)

| | | Parametric effect (percentile change of treated areas due to treatment) | Percentile change of treated areas, 2008-2014 | Percentile change of control areas, 2008-2014 |
|---|---|---|---|---|
| Engineering & Technical | *Engineering & Technology* | 5.7 | 1.7 | -3.0 |
| | *Design* | 5.4 | 4.2 | 0.5 |
| | *Building* | 2.5 | -8.4 | -9.5 |
| | *Mechanical* | 2.0 | -4.0 | -4.2 |
| | *Drafting* | 4.0 | -4.5 | -6.5 |
| | *Estimating* | 4.3 | 9.1 | 5.2 |
| Science | *Physics* | 3.8 | 0.1 | -3.6 |
| | Biology | 0.5 | 10.3 | 6.4 |
| Operation Management | *System Analysis* | 2.0 | 10.8 | 7.2 |
| | System Evaluation | 1.5 | 9.1 | 6.2 |
| | *Updating Knowledge* | 2.2 | 6.4 | 3.8 |
| | *Consulting* | 2.5 | 12.3 | 9.5 |
| Monitoring | Law | 0.5 | 7.8 | 5.1 |
| | *Compliance* | 4.8 | 11.2 | 6.2 |
| Residual | Geography | 0.2 | 6.2 | 4.7 |
| | Transport | 0.2 | -3.6 | -3.5 |

Results refer to coefficients estimated in Table E2**Error! Reference source not found.** based on equation 4. *GGS* labels in Italics denote a statistically significant (p<0.1) effect of environmental regulation.

Table E2 - Estimates by *GGS* item

| Engineering and Technical | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Item | NA in t=0 x trend | | NA designation | | NA implementation | | NA designation + NA implementation | | R sq | N |
| Engineering & technology | -0.000204 | (0.000167) | 0.00190*** | (0.000636) | 0.00102 | (0.000682) | 0.00291*** | (0.000676) | 0.389 | 4806 |
| Design | -0.000260 | (0.000164) | 0.00214*** | (0.000696) | 0.000871 | (0.000782) | 0.00301*** | (0.000772) | 0.321 | 4797 |
| Building | -0.0000498 | (0.000180) | 0.00110 | (0.000717) | 0.000350 | (0.000668) | 0.00145* | (0.000748) | 0.623 | 4815 |
| Mechanical | -0.000147 | (0.000195) | 0.00171* | (0.000894) | 0.000410 | (0.00106) | 0.00212* | (0.00108) | 0.504 | 4815 |
| Drafting | -0.000203 | (0.000146) | 0.000988 | (0.000666) | 0.000792 | (0.000748) | 0.00178** | (0.000734) | 0.552 | 4824 |
| Estimating | -0.0001 | -0.000115 | 0.00110** | -0.000463 | 0.000286 | -0.000479 | 0.00138*** | (0.000454) | 0.612 | 4815 |
| Science | | | | | | | | | | |
| Item | NA in t=0 x trend | | NA designation | | NA implementation | | NA designation + NA implementation | | R sq | N |
| Physics | -0.000164 | (0.000118) | 0.000842 | (0.000559) | 0.000534 | (0.000621) | 0.00138** | (0.000565) | 0.476 | 4824 |
| Biology | -0.0000479 | (0.0000782) | -0.000234 | (0.000386) | 0.000331 | (0.000358) | 0.0000972 | (0.000339) | 0.364 | 4824 |
| Operation Management | | | | | | | | | | |
| Item | NA in t=0 x trend | | NA designation | | NA implementation | | NA designation + NA implementation | | R sq | N |
| System analysis | 0.0000211 | (0.000119) | 0.000258 | (0.000482) | 0.000799 | (0.000524) | 0.00106* | (0.000565) | 0.621 | 4797 |
| System evaluation | 0.0000216 | (0.000129) | 0.000192 | (0.000525) | 0.000834 | (0.000553) | 0.00103 | (0.000634) | 0.606 | 4797 |
| Updating knowledge | -0.0000122 | (0.000100) | 0.000420 | (0.000482) | 0.000883 | (0.000561) | 0.00130** | (0.000556) | 0.446 | 4815 |
| Consulting | 0.000062 | -0.000135 | 0.000416 | -0.000504 | 0.000917** | -0.000403 | 0.00133** | (0.00056) | 0.631 | 4797 |
| Monitoring | | | | | | | | | | |
| Item | NA in t=0 x trend | | NA designation | | NA implementation | | NA designation + NA implementation | | R sq | N |
| Law | 0.000140 | (0.000135) | -0.0000827 | (0.000610) | 0.000394 | (0.000555) | 0.000311 | (0.000680) | 0.521 | 4824 |
| Compliance | -0.000205 | (0.000137) | 0.00167*** | (0.000492) | 0.0000999 | (0.000538) | 0.00177*** | (0.000558) | 0.446 | 4824 |
| Residual | | | | | | | | | | |
| Item | NA in t=0 x trend | | NA designation | | NA implementation | | NA designation + NA implementation | | R sq | N |
| Geography | 0.000145 | (0.000118) | -0.000481 | (0.000492) | 0.000156 | (0.000394) | -0.000324 | (0.000410) | 0.359 | 4824 |
| Transport | 0.000106 | (0.000211) | 0.000723 | (0.00111) | -0.00105 | (0.00115) | -0.000328 | (0.00127) | 0.466 | 4815 |

Dependent variables refer to SOC codes (2-digit) with at least one green job (SOC codes: 11, 13, 15, 17, 19, 23, 27, 41, 43, 47, 49, 51, 53). Fixed effect model weighted by kernel-based weights based on item-specific propensity score. Other control variables: state-specific year dummies; other controls interacted with linear trend: share of manufacturing (2005), share of primary sector (2005), share of construction sector (2005), share of utility sector (2005), import penetration (2005), log of population density (2005), log of average establishment size (2005), exposure to the crisis (2005).

Table E3 - Estimates for all occupations

| Panel A - *GGS* importance | | | | |
|---|---|---|---|---|
| | Science | Engineering & technical | Operation management | Monitoring |
| NA in t=0 x trend | 0.0000718 | -0.000127 | 0.00000640 | -0.0000400 |
| | (0.0000793) | (0.000123) | (0.000102) | (0.0000874) |
| NA designation | -0.000293 | 0.00124*** | 0.000187 | 0.000568 |
| | (0.000331) | (0.000407) | (0.000414) | (0.000420) |
| NA implementation | 0.000435 | 0.000372 | 0.000581 | 0.000155 |
| | (0.000300) | (0.000447) | (0.000404) | (0.000400) |
| NA designation + NA implementation | 0.000141 | 0.00162*** | 0.000768* | 0.000723** |
| | (0.000314) | (0.000507) | (0.000442) | (0.000364) |
| R sq | 0.473 | 0.574 | 0.593 | 0.603 |
| N | 4824 | 4806 | 4797 | 4824 |
| Panel B - Standard human capital measures | | | | |
| | RTI | log(training) | log(education) | Share requiring master degree |
| NA in t=0 x trend | -0.000123 | 0.000614 | 0.0000369 | 0.000185 |
| | (0.000234) | (0.000469) | (0.000111) | (0.000130) |
| NA designation | 0.000600 | -0.00203 | -0.000153 | -0.000774* |
| | (0.00101) | (0.00204) | (0.000459) | (0.000451) |
| NA implementation | -0.00161 | 0.00245 | 0.000866* | 0.000580 |
| | (0.00101) | (0.00210) | (0.000465) | (0.000441) |
| NA designation + NA implementation | -0.00101 | 0.000413 | 0.000713 | -0.000194 |
| | (0.00103) | (0.00179) | (0.000497) | (0.000435) |
| R sq | 0.631 | 0.300 | 0.594 | 0.640 |
| N | 4824 | 4815 | 4824 | 4815 |

Fixed effect model weighted by kernel-based weights based on propensity score. Dependent variables refer all SOC codes. Other control variables: state-specific year dummies; other controls interacted with linear trend: share of manufacturing (2005), share of primary sector (2005), share of construction sector (2005), share of utility sector (2005), import penetration (2005), log of population density (2005), log of average establishment size (2005), exposure to the crisis (2005).

Table E4 - Quantification of the effect of environmental regulation on *GGS* importance (only 2-digit SOC occupations with both brown and green jobs)

| | Parametric effect (percentile change of treated areas due to treatment) | Percentile change of treated areas, 2008-2014 | Percentile change of control areas, 2008-2014 |
|---|---|---|---|
| *Engineering & technical* | 2.9 | -0.7 | -2.4 |
| Science | 0.7 | 4.8 | 1.9 |
| Operation & Management | 1.0 | 7.4 | 4.8 |
| Monitoring | 1.2 | 5.8 | 3.8 |
| *Compliance* | 8.2 | 13.1 | 4.8 |

Results refer to coefficients estimated by means of equation 4 only for 2-digit SOC groups that contain both green and brown jobs. *GGS* labels in Italics denote a statistically significant ($p<0.1$) effect of environmental regulation.

Table E5 - Quantification of the effect of environmental regulation on *GGS* importance (only 3-digit SOC occupations with at least one green jobs)

| | Parametric effect (percentile change of treated areas due to treatment) | Percentile change of treated areas, 2008-2014 | Percentile change of control areas, 2008-2014 |
|---|---|---|---|
| *Engineering & technical* | 3.9 | -5.3 | -7.6 |
| Science | 1.7 | 0.9 | -1.6 |
| *Operation & Management* | 2.0 | 9.7 | 6.9 |
| *Monitoring* | 2.8 | 8.7 | 5.3 |

Results refer to coefficients estimated by means of equation 4 only for 3-digit SOC groups that contain at least one green jobs. *GGS* labels in Italics denote a statistically significant (p<0.1) effect of environmental regulation.

Table E6 - Quantification of the effect of environmental regulation on *GGS* importance for an alternative definition of regulatory change

| | Parametric effect (percentile change of treated areas due to treatment) | Percentile change of treated areas, 2008-2014 | Percentile change of control areas, 2008-2014 |
|---|---|---|---|
| *Engineering & technical* | 3.0 | -4.2 | -5.1 |
| *Science* | 2.7 | 1.5 | 1.7 |
| *Operation & Management* | 1.6 | 8.0 | 6.3 |
| Monitoring | 0.2 | 7.4 | 7.0 |
| *Compliance* | 2.5 | 6.7 | 8.2 |

Results refer to coefficients estimated by means of equation in footnote 28 that employs the alternative definition of regulatory change (see section 3.4). *GGS* labels in Italics denote a statistically significant (p<0.1) effect of environmental regulation.

Table E7 - Time-varying effect of environmental regulation on *GGS* importance

| | Engineering & technical | Science | Operation management | Monitoring | Compliance |
|---|---|---|---|---|---|
| NA in t=0 x trend | -0.000208 | -0.000101 | 0.00000407 | -0.0000239 | -0.000185 |
| | (0.000144) | (0.0000806) | (0.000110) | (0.000107) | (0.000132) |
| 2 years before NA switch | -0.00125 | -0.000430 | 0.0000504 | -0.000226 | -0.000468 |
| | (0.000848) | (0.000504) | (0.000521) | (0.000498) | (0.000607) |
| 1 year before NA switch | -0.000543 | -0.000356 | -0.0000808 | -0.0000487 | 0.000112 |
| | (0.000548) | (0.000411) | (0.000348) | (0.000381) | (0.000448) |
| Na switch | (reference year) | (reference year) | (reference year) | (reference year) | (reference year) |
| 1 year after NA switch | 0.00103** | 0.000197 | 0.000532 | 0.000996** | 0.00179*** |
| | (0.000506) | (0.000285) | (0.000386) | (0.000386) | (0.000404) |
| 2 years after NA switch | 0.00105* | 0.000240 | 0.000551 | 0.000639 | 0.00163*** |
| | (0.000554) | (0.000335) | (0.000484) | (0.000491) | (0.000577) |
| 3 years after NA switch | 0.00194*** | 0.000765* | 0.000739 | 0.000697 | 0.00169*** |
| | (0.000669) | (0.000407) | (0.000608) | (0.000511) | (0.000588) |
| 4 years after NA switch | 0.00149** | 0.000483 | 0.00173*** | 0.00111** | 0.00158** |
| | (0.000727) | (0.000426) | (0.000638) | (0.000514) | (0.000647) |
| 5 years after NA switch | 0.00129 | -0.000233 | 0.000717 | -0.000419 | 0.000435 |
| | (0.00110) | (0.000651) | (0.000931) | (0.000894) | (0.000975) |
| 6 years after NA switch | 0.00102 | -0.000410 | 0.000665 | -0.000784 | -0.000285 |
| | (0.00130) | (0.000822) | (0.00126) | (0.00118) | (0.00126) |
| R sq | 0.601 | 0.391 | 0.615 | 0.552 | 0.448 |
| N | 4824 | 4824 | 4815 | 4815 | 4824 |

Dependent variables refer to SOC codes (2-digit) with at least one green job (SOC codes: 11, 13, 15, 17, 19, 23, 27, 41, 43, 47, 49, 51, 53). Fixed effect model weighted by kernel-based weights based on propensity score. Other control variables: state-specific year dummies; other controls interacted with linear trend: share of manufacturing (2005), share of primary sector (2005), share of construction sector (2005), share of utility sector (2005), import penetration (2005), log of population density (2005), log of average establishment size (2005), exposure to the crisis (2005).